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From Instant Cues to Fading Views: Piloting Just-in-Time and Fading Strategies in Digital Habit Formation

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From Instant Cues to Fading Views: Piloting Just-in-Time and Fading Strategies in Digital Habit Formation

Completed Research Full Paper

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Abstract

Non-communicable diseases impose a global burden, with physical inactivity as a key modifiable risk factor. Digital interventions can initiate behavior change, yet sustaining these changes is challenging. This pilot study evaluates two reminder strategies—Just-in-Time (JIT) and fading reminders—in fostering habit formation among 18 participants randomized into four groups (JIT, two fading conditions, and a control) using a mobile app to track morning exercise routines. Results reveal that JIT reminders significantly boost initial habit adoption, while fading reminders promote greater habit strength. A structured, gradual transition between reminder types appears critical for long-term habit maintenance and automaticity. Furthermore, this study contributes a research model that integrates reminder strategies, habit strength, and behavioral outcomes, offering a theoretical framework for designing tailored digital interventions to reinforce lasting behavioral change as part of digitally transformed, patient-centered healthcare.

Keywords

Digital Health Intervention, Habit Formation, Reminder, Behavior Change

Introduction

Non-communicable diseases (NCDs), including cardiovascular diseases, diabetes, and cancer, represent a global health burden, accounting for approximately 74% of all deaths worldwide (WHO, 2022). These diseases are mainly driven by modifiable risk factors such as physical inactivity and unhealthy diets (Budreviciute et al., 2020). Promoting health-enhancing behaviors is, therefore, a key priority in public health (WHO, 2013). While digital interventions have demonstrated short-term success in promoting health-enhancing behavior, the challenge remains in ensuring long-term sustainability (Roth et al., 2023; Rupp et al., 2024). The formation of habits is discussed for achieving lasting behavior changes, as habits have been shown to impact long-term behavior (Cleo et al., 2020). Habits are automatic behaviors that develop through repetition in the same context (Lally & Gardner, 2013; Verplanken, 2018; Wood & Neal, 2007). For example, brushing one's teeth (behavior) before bed (context) typically becomes a habit through consistent repetition, reducing conscious decision-making over time. However, habit formation is not linear and requires perseverance (Wang et al., 2020). Prior research suggests digital strategies for habit formation, emphasizing the potential of reminders (Pinder et al., 2018; Stark et al., 2025). Reminders can prompt behavior repetition and strengthen the mental link between behavior and context, essential for recalling the behavior when the context arises. Different types of reminders have been discussed for their roles in facilitating behavior repetition or strengthening the behavior-context link. Just-in-time (JIT) reminders

provide prompts when the context typically arises, the moment an individual is most likely to engage in the behavior, making JIT reminders particularly useful in the early stages of habit formation when consistency is key (Stawarz et al., 2015). In contrast, fading reminders gradually reduce over time, encouraging individuals to strengthen the behavior-context link, which could help them remember the habit even without reminders (Nobbe et al., 2024). While theoretical frameworks suggest that reminders should be gradually removed to ensure self-sustaining habits, empirical evidence is lacking (Pinder et al., 2018). This study, therefore, investigates the two forms of reminders—JIT and fading—and their contribution to habit formation. Specifically, it examines the following research question (RQ):

RQ: How do Just-in-Time and fading reminders influence the formation of habits?

To address the RQ, we conducted a pilot experiment with four groups: one JIT group, two fading groups with different fade-out times, and a control group without reminders. Our findings suggest that JIT reminders support habit execution in the early stages, while a well-timed transition to fading reminders is crucial for long-term adherence. Fading led to higher dropout rates, indicating the need for habit refinement. Additionally, a structured, gradual transition rather than an abrupt switch may strengthen the habits and enhance self-reported automaticity. Our contribution is threefold. First, we provide empirical insights on the role of different reminder strategies in habit formation. Second, we highlight the importance of refining habits in the early stages to prevent disengagement, and third, we offer insights into optimizing the transition between JIT and fading reminders to maximize long-term adherence.

Theoretical Background and Hypothesis Development

A habit is a learned behavior that, through repetition in a stable context, becomes an automatic response requiring minimal cognitive effort to execute (Lally & Gardner, 2013; Verplanken, 2018). Unlike goal-directed behaviors, habits rely on context-behavior associations rather than conscious decision-making and can persist even when motivation fluctuates, making them a powerful tool for long-term behavior change. The habit formation model (Figure 1) conceptualizes the habit formation process as a progression through four stages: Intention, Action, Repetition, and Automaticity (Lally et al., 2010; Lally & Gardner, 2013).



Figure 1. Four Stages of Habit Formation adapted from (Lally et al. 2010; Lally and Gardner 2013)

During the Intention stage, individuals decide to adopt a behavior in a specific context, typically driven by a health goal or external motivation. The Action stage marks the first execution of the behavior, which is then entirely intentional and requires commitment. This initial execution helps form a link between behavior and context. Furthermore, adjustments are recommended if needed to ensure fit. In the Repetition stage, the link is strengthened. This phase marks the beginning of the transition towards automaticity, with habit strength increasing rapidly at first, then stabilizing over time (Lally et al., 2010). In the Automaticity stage, the behavior becomes habitual and is performed with minimal conscious effort.

Without structured support, nowadays often delivered digitally, many individuals fail to achieve automaticity (Sheeran & Webb, 2016). Digital interventions, particularly Just-in-Time Adaptive Interventions (JITAIs), are designed to provide reminders when individuals are most receptive to engaging in the behavior (Nahum-Shani et al., 2018). Prior research has explored the role of reminders in digital behavior change interventions, with a focus on JIT reminders. Studies have investigated how such reminders can support medication adherence and examined users' experiences with reminders in habit formation tools (Renfree et al., 2016; Stawarz et al., 2015). Yet, prior research focused on the sole act of reminding but not on comparing reminder types. During habit formation, reminders serve different functions, such as promoting repetition, reinforcing the behavior-context link, or gradually fading out, making the interplay and timing of varying reminder types relevant. This approach involves adjusting the interval between reminders, either keeping it constant or progressively increasing it over time. Notably, this gap is also apparent in most contemporary behavior change frameworks, which largely neglect timing (Pelly et al., 2023). For habit formation, JIT reminders are especially suited in the Repetition stage, as they

support habit execution despite fluctuations in motivation. JIT reminders provide consistent, timely prompts that help mitigate these fluctuations, ensuring that individuals regularly execute the behavior and may be more suited than fading reminders, which do not steadily deliver the prompt at the time the habit is supposed to be executed (Nobbe et al., 2024). Based on this theoretical foundation, we hypothesize that:

H1a: Just-in-time reminders lead to higher habit completion rates than fading reminders during the reminding phase.

As the habit strengthens, excessive reliance on external reminders may emerge, which can hinder the transition to autonomous behavior (Stawarz et al., 2015). Prior research has suggested that reminders should be discontinued once a certain level of repetition is reached to avoid over-reliance on reminders and facilitate the transition toward automaticity (Nobbe et al., 2024). Therefore, we hypothesize that:

H1b: After an initial reminding phase, fading reminders lead to higher habit completion rates than just-in-time reminders.

Despite the argument that fading reminders can help prevent over-reliance on reminders, they also pose the risk of insufficient engagement. Without consistent exposure to reminders, as seen with JIT reminders, individuals may receive fewer signals reinforcing the behavior, potentially leading to a decline in motivation and an increased likelihood of disengagement (Stawarz et al., 2015). Therefore, we hypothesize that:

H1c: Fading reminders lead to a higher dropout rate than just-in-time reminders.

The habit completion rate may not be the most relevant outcome. For a habit to become strong, habit strength is essential. Habit strength reflects the degree of automaticity a habit attains as perceived by the individual and is assessed using the Self-Report Habit Index (SRHI). Regarding habit strength, fading reminders have been discussed as being most effective, as they are intended to prevent over-reliance on reminders and encourage the individual to recall the habit on their own (Nobbe et al., 2024; Renfree et al., 2016). Therefore, we hypothesize that:

H2: Fading reminders lead to a higher SRHI than just-in-time reminders.

As individuals develop stronger habits, their cognitive effort decreases, and they begin performing the behavior automatically. With a strong habit, behavior no longer relies on reminders and is consistently executed whenever the context arises. Thus, a strong habit, which a high SRHI characterizes, is expected to have a high completion rate. This leads to the following hypothesis:

H3: A higher SRHI leads to a higher habit completion rate.

Figure 2 presents the constructs. Reminder types serve as the independent variable. Habit strength (measured by SRHI) reflects the user's perceptions (psychological, cognitive-affective), while dropout rate and habit completion represent behavioral outcomes. The colors align with those of the habit formation stages in Figure 1.

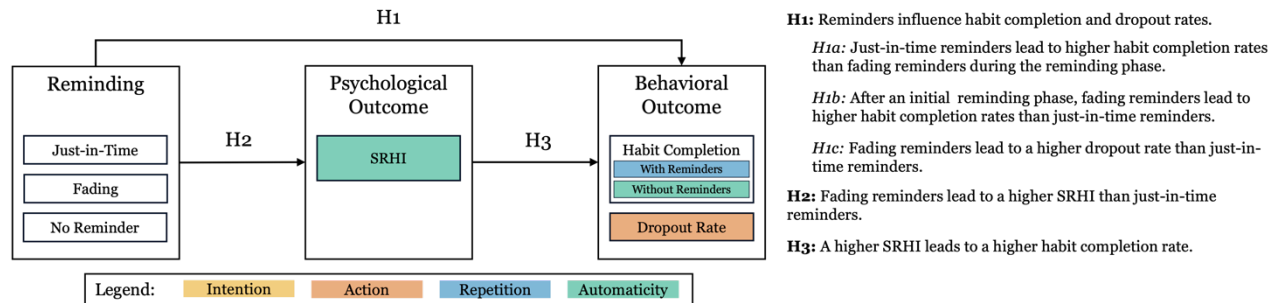


Figure 2. Research Model and Hypotheses

Method

This study pilots an experiment investigating the effects of different reminders on habit formation. This pilot initially tests and refines hypotheses, explores methodological approaches, and identifies key factors

for a larger-scale future study. The study utilized HabHealth, a mobile application designed to facilitate habit formation. While HabHealth provided the platform for delivering reminders and collecting behavioral data, it functioned primarily as a research tool rather than the focal point of the study. Participants used the app to set habit-related goals, receive scheduled reminders, and complete surveys throughout the study.

Recruitment took place at a German University from Sept. to Nov. 2024 through flyer distribution at the university, gyms, workplaces, and personal outreach via a study newsletter. Individuals subscribed to the newsletter and received periodic study updates. Once the HabHealth application was operational, potential participants were invited to join the study and download the app. Thirty-nine individuals initially subscribed, with 18 meeting the inclusion criteria (aged 18 or older) and completing the baseline survey. Participants were randomly assigned to one of four experimental groups. The final sample comprised 10 men and eight women aged 18 to 64. In terms of educational background, one participant held a bachelor's degree, 11 had either a master's or doctoral degree, while the remaining participants had vocational training ($n = 1$), apprenticeship training ($n = 1$), or were still in undergraduate education ($n = 4$). Regarding occupational status, one participant was unemployed, five were students, and the remaining 12 were employed full-time or part-time. The study focused on forming a morning exercise habit, requiring to engage in a brief physical activity each morning. Participants selected their preferred exercise (e.g., yoga, stretching, bodyweight workouts) and specified an execution time between 4 and 11 AM. We deliberately allowed participants to choose the type, duration, and intensity of their morning exercise to maximize relevance and reduce barriers to adherence, especially given the expected dropout in this pilot study. The study design is summarized in Figure 3.

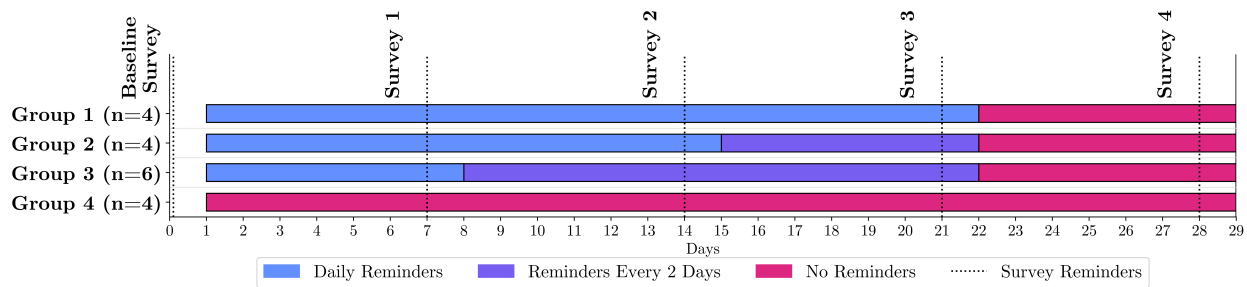


Figure 3. Study Design

Reminders were discontinued in the fourth week to assess habit strength. Reminders followed a standardized format to reinforce participants' habit formulation (e.g., "Whenever I get up in the morning, I do my 10-minute morning exercise"). Data collection included habit completion tracking via the app and five online surveys via LimeSurvey, conducted at baseline and weekly. These surveys evaluated psychological (e.g., SRHI) and behavioral aspects (e.g., habit completion). Participants recorded their habit completion within the app daily, reporting whether they had engaged in their designated habit.

Results

Our analyses were conducted in Python using SciPy. Figure 4 depicts the number of completed habits per participant (HCPP), mean SRHI scores at each survey time point, and the dropout rates by groups. The dropout rate, based on survey completion, varied across the experimental groups. Group 1 demonstrated full participation throughout the study. Groups 2 and 3 experienced dropout rates of 75% and 83.3%, indicating greater dropout in the fading reminder conditions. Group 4 had a dropout rate of 25%. Overall, 9 (50%) out of the initial 18 participants completed all surveys. This result falls within the confidence intervals (29–57%) for comparable app-based interventions (Meyerowitz-Katz et al., 2020). HCPP values ranged from 0 (no remaining participants completed the habit on a given day) to 1 (all remaining participants did), revealing distinct behavioral patterns. Group 1 maintained a high HCPP during the reminder phase and a decline once reminders ceased. Group 2 showed initial fluctuation but maintained an upward trajectory, similar to Group 3. Group 4 (control group) exhibited the highest day-to-day variability without a trend. Statistical analysis indicated a significant group difference in habit completion during the first 21 days ($p < 0.001$), with Group 1 showing the highest habit completion. Once reminders

were discontinued, this difference was no longer statistically significant ($p=0.0783$). The groups exhibited distinct patterns regarding habit strength (depicted as dots in Figure 4). Group 1 reached its highest SRHI scores in Week 3, followed by a decline in Week 4, aligning with the observed drop in HCPP. Groups 2 and 3 steadily increased SRHI scores throughout the study, suggesting gradual habit strengthening. Group 4 consistently reported low habit strength. Despite these differences, statistical analysis revealed no significant group differences for SRHI scores in week 4 ($p=0.294$) despite observable trends.

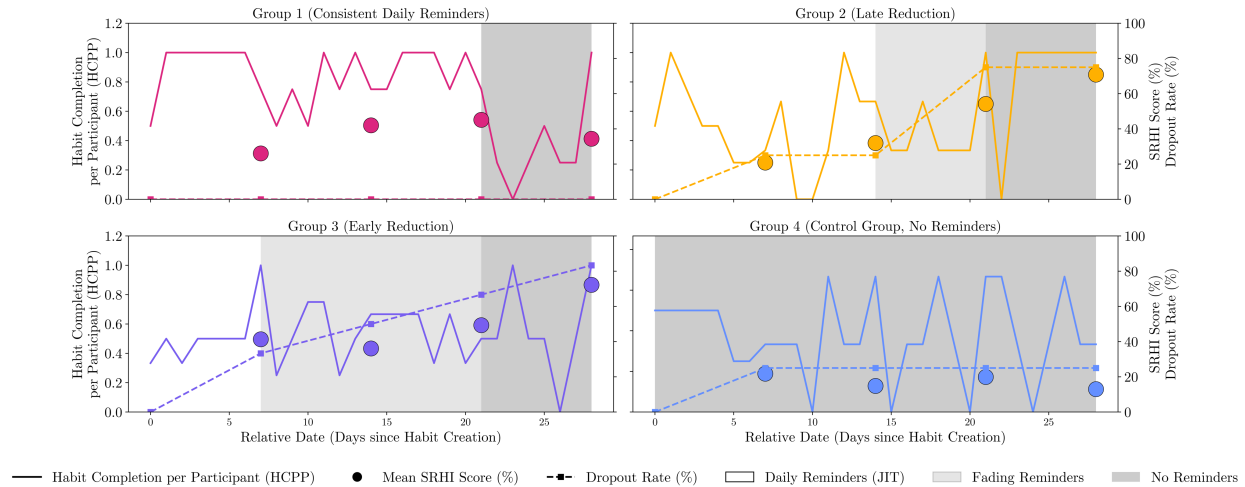


Figure 4. Completed Habits per Participant, SRHI-Score, and Dropout Rate by Group

At baseline, users completed a survey assessing factors potentially relevant to habit formation (Figure 5), including physical activity levels (Global Physical Activity Questionnaire), well-being (WHO-5), stress overload (SOS-S), and obsessive-compulsive traits (OCI-R) (Amirkhan, 2018; WHO, 2021, 2024). The Ten-Item Personality Inventory (TIPI) was used to assess baseline personality traits, including Extraversion (Ext), Agreeableness (Agr), Conscientiousness (Con), Emotional Stability (ES), and Openness to Experience (Ope) (Gosling et al., 2003; Muck et al., 2007). While variations were observed, no significant differences emerged between the groups. The Kruskal-Wallis Test confirmed that traits were comparable across conditions (all $p > 0.16$), suggesting that personality differences did not introduce confounding effects.

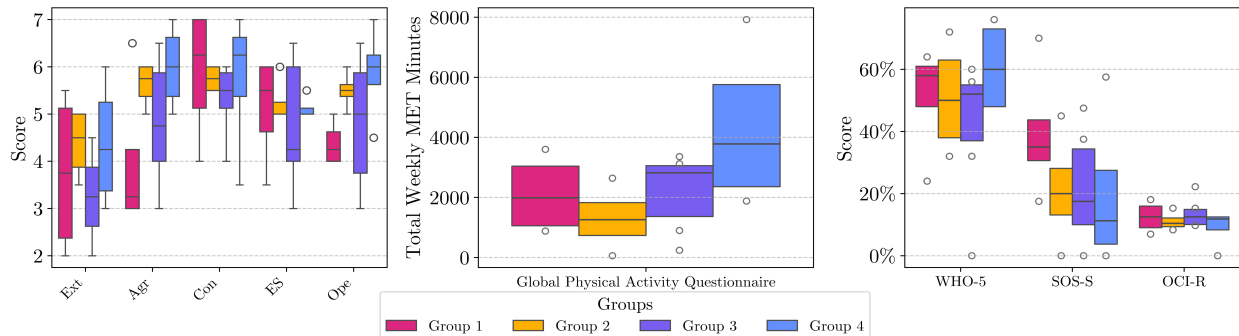


Figure 5. TIPI-, GPAQ-, WHO-5-, SOS-S-, OCI-R-Results by Group

The Global Physical Activity Questionnaire (GPAQ) revealed variation in weekly MET minutes (Metabolic Equivalent of Task, a measure of energy expenditure). Group 4 reported the highest physical activity levels. Groups 2 and 3 exhibited lower baseline activity (WHO, 2021). However, no statistically significant differences between groups were found ($p > 0.29$). The WHO-5 Well-Being Index showed minor fluctuations, with nearly all median values ranging between 50% and 60%, and no significant differences detected ($p > 0.63$; WHO, 2024). The Short Stress Overload Scale (SOS-S) indicated that Group 1 reported the highest initial stress levels and Group 4 the lowest. Still, no significant group differences ($p > 0.55$) were found. Similarly, the Short Obsessive-Compulsive Inventory (OCI-R) showed minor baseline differences,

with Group 3 reporting the highest scores, yet no significant differences found across groups ($p > 0.80$). To better understand habit formation, we analyzed correlations between baseline and week 4 factors. Figure 6 presents correlation coefficients between baseline variables and the key outcomes. Among the personality traits (Ext, Agr, Con, ES, Ope), Openness to Experience (Ope) and Agreeableness (Agr) exhibit the most substantial negative correlations with habit completion, whereas Extraversion (Ext), Conscientiousness (Con), and Emotional Stability (ES) show weaker associations. Regarding SRHI scores, Con demonstrates the strongest negative correlation among all observed variables, while Ext and Agr show moderate positive correlations. For physical activity measures, total MET minutes per week correlate positively with habit completion and SRHI scores. In contrast, sedentary MET (Sed MET) negatively correlates with both outcomes. Stress (SOS-S) demonstrates a modest negative relationship with habit completion and SRHI. Obsessive-compulsive traits (OCI-R) are negatively associated with SRHI scores but have a negligible correlation with habit completion.

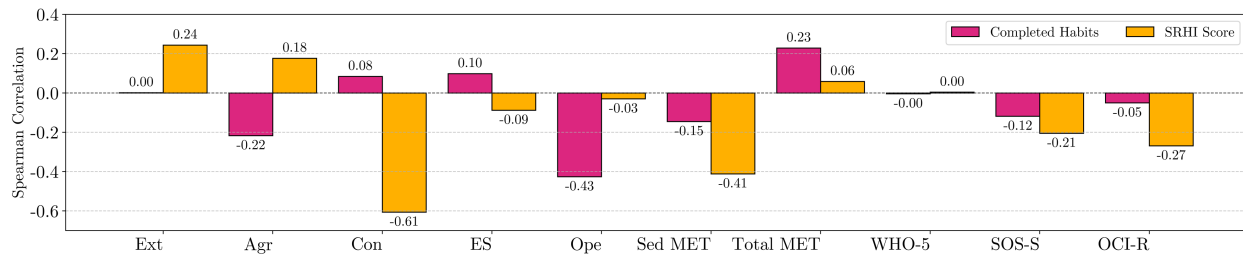


Figure 6. Correlation Coefficients with the Number of Completed Habits and SRHI Score

Table 1 summarizes the hypothesis testing. The results indicate that JIT reminders led to the highest habit completion rates (H1a), while no significant differences were found after reminders ceased (H1b). Dropout rates were significantly higher in the fading reminder groups (H1c), no group differences in SRHI were observed (H2), and higher SRHI scores led to significantly higher habit completion rates (H3).

Hypothesis	Observed Metric	Statistical Test	p-value	Description
H1a	Group Difference for total Habit Completion with reminders.	Kruskal-Wallis with Bonferroni Correction ($\alpha=0.0250$)	1.9e-09 ***	The JIT-Reminder Group (Group 1) demonstrated the highest habit completion rate.
H1b	Group Difference for total Habit Completion without reminders.	Kruskal-Wallis with Bonferroni Correction ($\alpha=0.0250$)	0.078	Completed Habits decreased most but not significantly in Group 1.
H1c	Group Difference in dropout rates.	Chi-Square test of independence	0.034 *	Groups 2 and 3, which received fading reminders, exhibited a higher dropout rate.
H2	Group Differences in SRHI score (Week 4).	Kruskal-Wallis	0.294	Participants in the fading reminder groups reported higher (but not significantly higher) SRHI scores at the end of the study.
H3	Correlation between SRHI score (Week 4) and total number of habit completions per participant.	Spearman Correlation Test (Correlation Coefficient: 0.9048)	0.002 **	Strong monotonic relationship between SRHI scores and habit completion.

Table 1. Statistical Summary (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

Discussion

This pilot study investigated the effects of two reminder types, JIT reminders and fading reminders with two different fading onset points, on habit formation. Habit formation progresses through four key stages: Intention, Action, Repetition, and Automaticity (see Figure 1). In the following, we present and interpret key findings along the four stages of the habit formation process.

Dropout as an Indicator for Habit Refinement (H1c)

This study showed that fading reminders led to a significantly higher dropout rate than JIT reminders. While dropout is often interpreted negatively, it can also provide valuable opportunities for habit refinement (Eysenbach, 2005; Renfree et al., 2016). In the early stages of habit formation, dropout may indicate a misalignment—either the chosen habit does not fit into the user’s routine (suboptimal context), or the selected behavior is not ideal (suboptimal behavior). The lower dropout rate in the JIT reminder group suggests that these reminders help users sustain engagement but might also reinforce habits that are not optimally designed for long-term adherence. A key implication of H1c is that dropout in the early stages of habit formation should not be viewed as a failure but rather as an indicator that refinement is needed. To prevent dropout from becoming final, digital interventions should incorporate churn prediction techniques that identify early disengagement risks and guide users into a habit refinement cycle. This suggests a need for a new type of reminder—such as a **re-engagement reminder**—that prompts users to reassess and adapt their habits before disengagement becomes permanent. Furthermore, this finding challenges the assumption that habit formation follows a strictly linear sequence through the four stages. Instead, early dropout can indicate the need for iteration between stages, particularly from the Action or Repetition stage back to the Intention stage. This iterative approach is particularly valuable in digital interventions, where reminders and feedback loops can help users adjust their habits dynamically, ensuring stronger long-term adherence.

JIT Reminders and Habit Execution Consistency (H1a)

Our findings also indicate that JIT reminders led to significantly higher habit completion rates compared to fading reminders in the reminder phase. This finding highlights the critical role of JIT reminders in the Action and early Repetition stages, where consistency in execution is essential. These reminders serve as immediate cues that reinforce behavior when habits are supposed to be executed, reducing the cognitive effort required for adherence. From a practical perspective, this finding suggests that incorporating a dedicated JIT reminder phase during habit formation is beneficial. Ensuring strong execution consistency before transitioning to fading reminders could help build a solid foundation for long-term adherence.

SRHI and Habit Stability (H3)

We also found a significant positive correlation between SRHI scores in Week 4 and habit completion rates. This finding suggests that completion rates increase as habits become more automatic. This is a crucial insight as higher habit completion rates enhance the likelihood of meaningful behavioral outcomes. Additionally, we observed a trend in H2, where participants in the fading reminder groups reported higher SRHI scores at the end of the study period. This finding suggests that fading reminders may be particularly beneficial in the Automaticity stage and that the transition from JIT to fading reminders should ideally occur between the Repetition and Automaticity stages to solidify long-term habit strength.

Optimizing the Transition from JIT to Fading Reminders

Although not hypothesizing an optimal transition point, our study allowed us to compare two different fading onset times (Group 2 vs. 3). Our results suggest that beginning the fading-out process too early (Group 3) leads to increased dropout, whereas delaying it slightly (Group 2) results in more sustained engagement. We also observed that Group 2 maintained a relatively stable habit completion rate once reminders were ceased, whereas Group 3 exhibited more variability. This could indicate that a later fading-out process might be the most effective approach.

Our findings lead to two further considerations. First, we propose a refined theoretical framework for habit formation in digitally supported environments, as depicted in Figure 7. In this model, the four stages are still sequential, as discussed initially, but allow for a refinement cycle to make the habit more fitting. Furthermore, we suggest applying different reminders for the various stages. JIT reminders may be beneficial during Action and early Repetition, while transitions should occur afterward to fading reminders. Furthermore, a new reminder would allow to fine-tune the habit to gain a better fit. In habit formation, disengagement often occurs when a behavior does not feel personally relevant or sustainable, leading to declining motivation and dropout (Lally & Gardner, 2013). The re-engagement reminder counteracts this by prompting the habits' reassessment, ensuring alignment with goals, and circumstances. Based on this rationale, we introduce Hypotheses H1d and H1e, examining the impact on initial and final dropout rates.

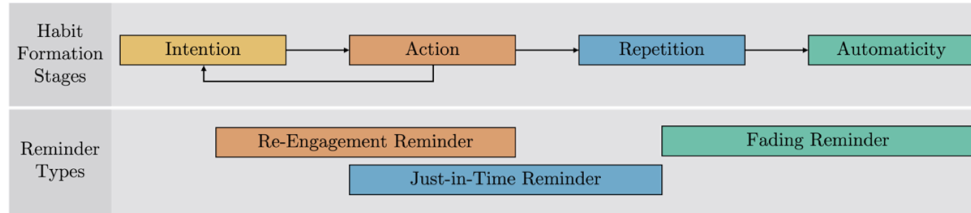


Figure 7. Adaptive Habit Reminder Model

Second, we propose a refined research model (Figure 8), including the two reminder types we investigated in this study and the additional one we propose to include. According to the previous discussion we assign JIT reminders to the Repetition stage, fading reminders to the Automaticity stage, and the re-engagement reminder to the Action stage.

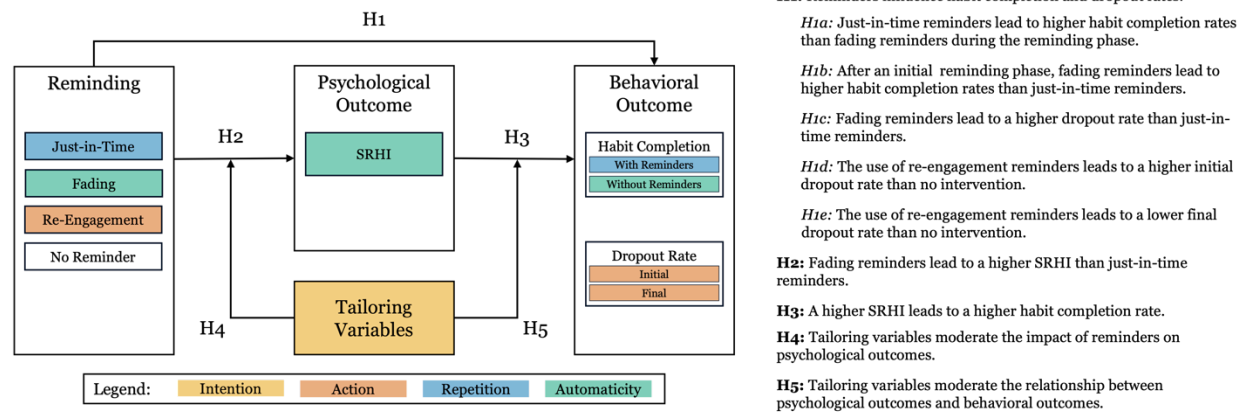


Figure 8. Revised Research Model

Furthermore, we introduce Tailoring variables as possible moderators with a potential effect on the relationship between Reminders and SRHI (H4) and Habit Completion (H5) (Reinsch et al., 2024). A candidate for examining H4 includes openness to experience, as indicated by the negative relation with habit completion. It may be cautiously interpreted that those more open to diverse kinds of experience may experience habits as a corset so that another type of intervention may be more beneficial. A candidate for H5 includes sedentary MET. As summarized in Figure 6, we found a negative correlation between sedentary MET and SRHI. If this relationship is confirmed, an implication may be to start with habits in the physical activity domain to reduce sedentary MET and hence lay the basis for implementing stronger habits.

Limitations

The present study provides valuable preliminary insights into the role of reminder strategies in habit formation. Yet, its scope is tempered by two interrelated limitations: the sample size and the study duration. As a pilot trial, our investigation was designed primarily to assess feasibility and explore early hypotheses. Consequently, the limited sample size constrains the generalizability of our findings and reduces the

statistical power. Our chi-square test showed significant differences in dropout rates between groups, but since some expected counts were less than 5, the results should be interpreted with caution. Compounding the issue of sample size is the study's four-week duration. Habit development is a complex process, with previous research suggesting longer habit formation periods (Lally et al., 2010). Due to our limited timeframe, it remains unclear whether the observed behaviors are merely transient snapshots or enduring habits that no longer require external reinforcement.

Future Research

While yielding significant preliminary findings, the present study was constrained by a limited sample size and a relatively short duration, both of which may have introduced challenges to the robustness of the outcomes. The follow-up study will employ a substantially larger cohort and an extended observation period. Based on an assumed medium effect size ($\eta^2 = 0.06$), a G*Power analysis indicated that a minimum of 93 participants per group would be requisite (Faul et al., 2007). Accounting for a dropout rate of 50%, as experienced in the pilot phase, 186 participants per group should be recruited. In addition, we plan to refine the control group design to reduce engagement-related confounds, introducing neutral, non-habit-related notifications to maintain a comparable level of app interaction. This adjustment will help ensure that differences in outcomes are more clearly attributable to the specific characteristic of the reminders rather than to general engagement effects. Beyond, to enhancing statistical robustness, the increased sample diversity will likely introduce greater variability in demographic and personality characteristics, presenting an intriguing avenue for future studies. Correlations between perceived and executed habit strength varied with personality traits, as illustrated in Figure 6, suggesting a potential bias in self-perception. Future research should elucidate the reasons for this divergence by exploring how personality traits influence responses on the SRHI questionnaire. Exploring the link between habit strength and personality traits would support the call for more personalized digital health interventions and facilitate the development of tailored strategies to improve healthcare outcomes (Cancela et al., 2021; Reinsch et al., 2024; Wang et al., 2020). Ultimately, providing more effective tools in the ongoing battle against NCDs.

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